

1. Motivation

27 Mt

Copper demand, 2024¹

21%

Heap leaching share of copper²

25%

Ore grade decline³

Demand for copper is driven by electrification and net-zero targets, while rich ores are being exhausted and grades have fallen 25% between 2003 and 2013. Announced mining projects are projected to meet only 70% of demand by 2035¹.

Heap leaching unlocks the low-grade ore that must fill this 30% gap. Recovering it efficiently is now a sustainability priority, not just a technical one.

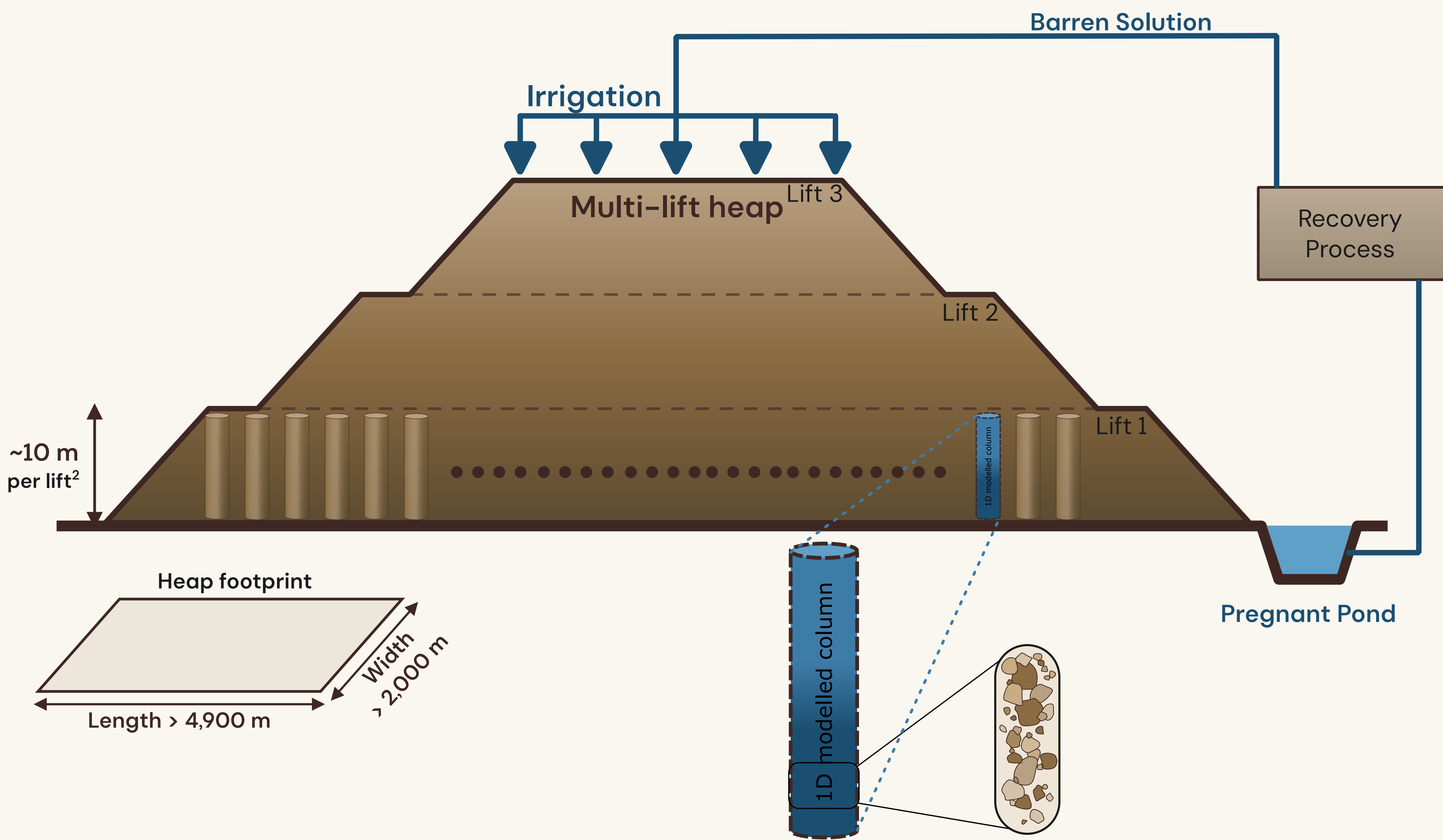


Figure 1: Copper heap leaching circuit. Lixiviant is irrigated over a multi-lift heap, percolates through the ore to dissolve copper, and is collected as pregnant leach solution before recovery and recycling. The heap is modelled as independent 1D columns, the control volume this work targets, with a zoom showing the heterogeneous packing that drives preferential flow.

2. Problem Statement

Copper recovery depends on how well the leaching solution wets the ore, which is set by irrigation. But heaps are heterogeneous, so solution channels through preferential paths and bypasses stagnant dry zones that leach poorly^{4,5}.

Recovery therefore hinges on the internal saturation state, yet operations still irrigate at a fixed, uniform rate that ignores it⁶.

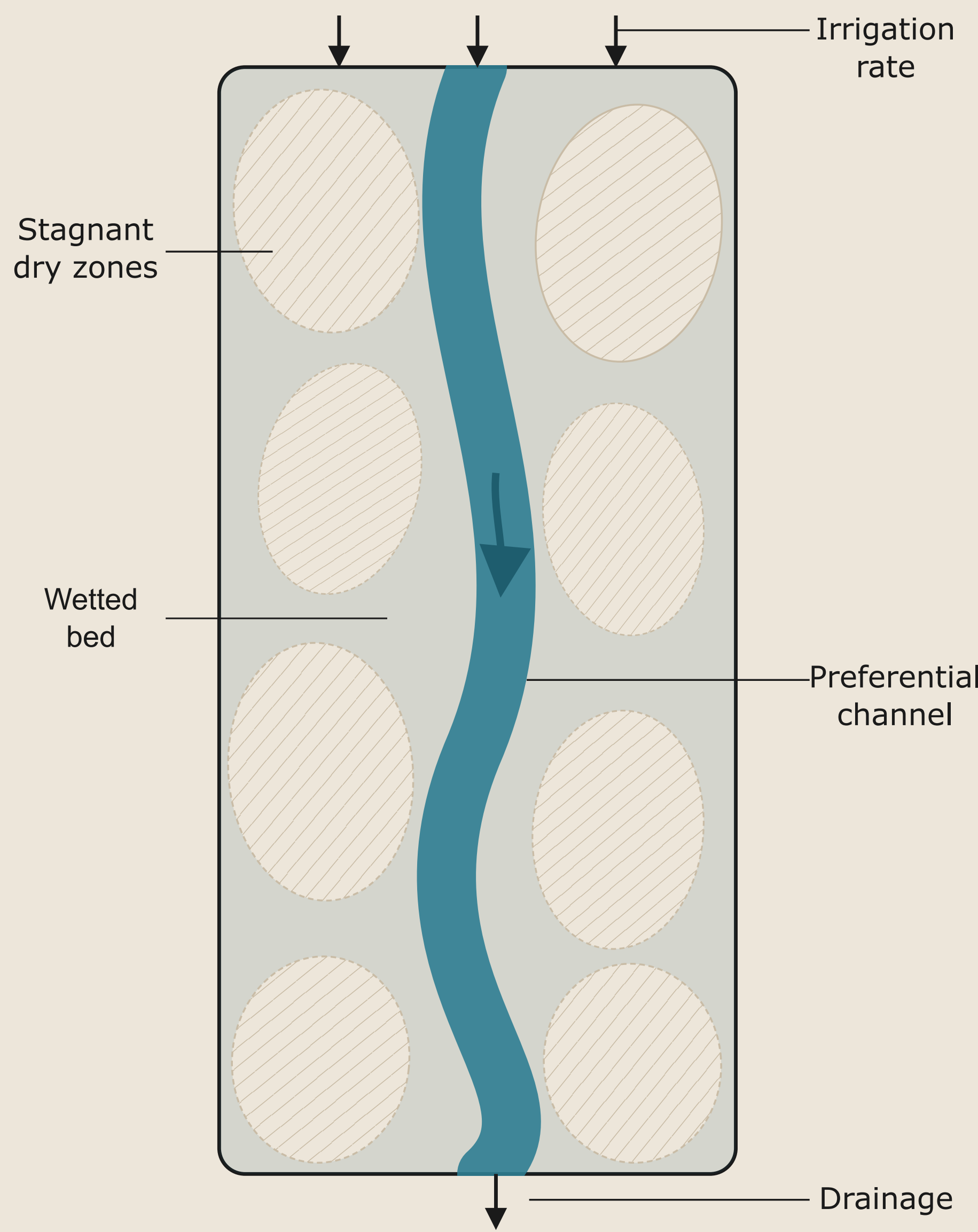


Figure 2: Preferential flow in an unsaturated ore column. Solution channels through wetted paths and bypasses dry stagnant zones that leach poorly.

5. Results

< 1 second

for a full leach cycle of 25 days per column

This replaces the fitting step that made real-time control impossible at heap scale.

A single fast evaluation now does the same job to enable real-time control. Measured on a single column.

6. Sustainability

Better control leads to more sustainable copper production from low-grade ore, using fewer resources

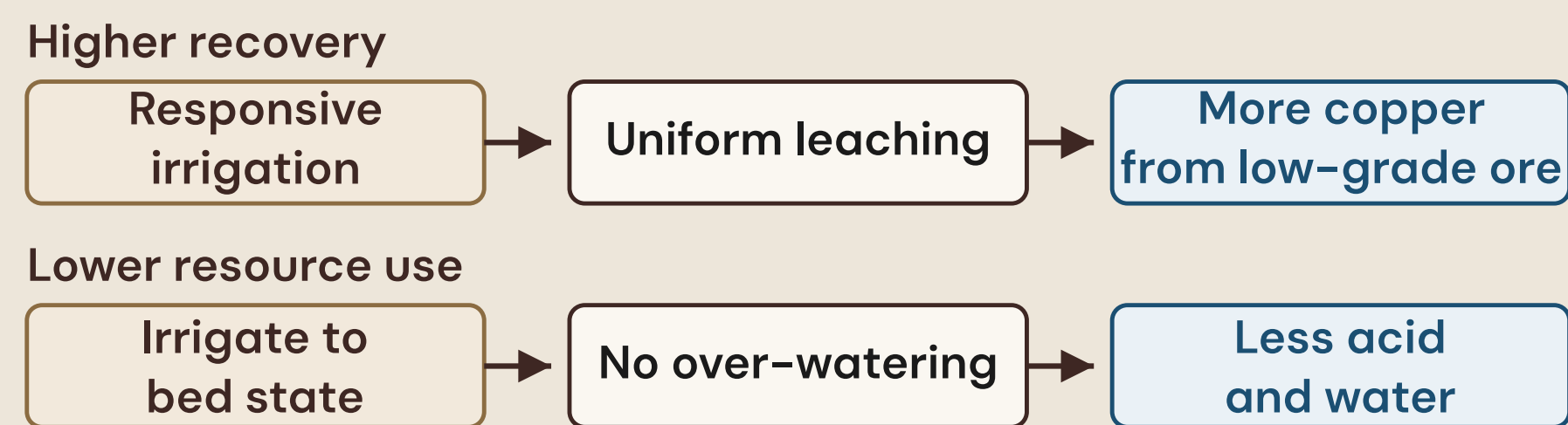


Figure 5: Sustainability impact pathways

7. Conclusion

This work shows a physics-guided surrogate can replace the fitting step that blocks real-time control across the full heap, running in a single fast evaluation while staying physically consistent.

4. Contribution

This work replaces the costly fitting step with a supervised machine learning (ML) model to predict heap saturation in a single fast evaluation.

The fitting step is not eliminated but relocated to training, done once beforehand instead of every control decision.

The model is physics-guided (PGML), a discretised Gardner residual is imposed in its training loss so it adheres to physics and stays consistent even with scarce data^{7,8}.

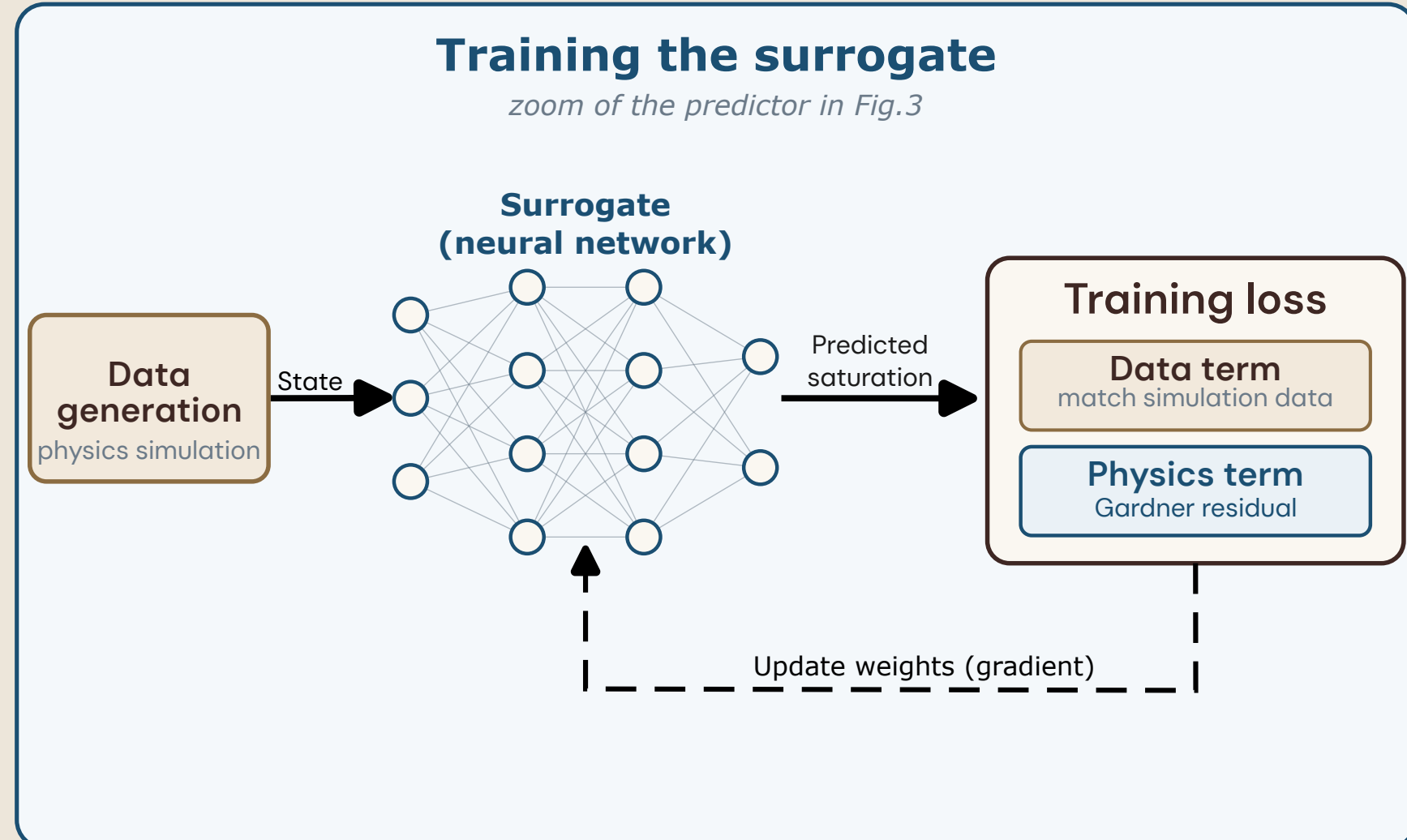


Figure 4: Training the surrogate. The network learns to predict saturation by minimising both a data error and a Gardner physics residual, keeping predictions physically consistent.

3. State of the Art

The state of the art is a nonlinear model predictive controller (NMPC)⁶ that adjusts irrigation in real time to reach a target saturation.

It splits the heap into many 1D columns and controls each one, using physics to predict how saturation responds.

Every control decision runs two calculations, one to fit the model to recent data and one to optimise irrigation.

The fitting step is the bottleneck, and its cost multiplies as the heap is split into more columns, so the method cannot run in real time at full scale.

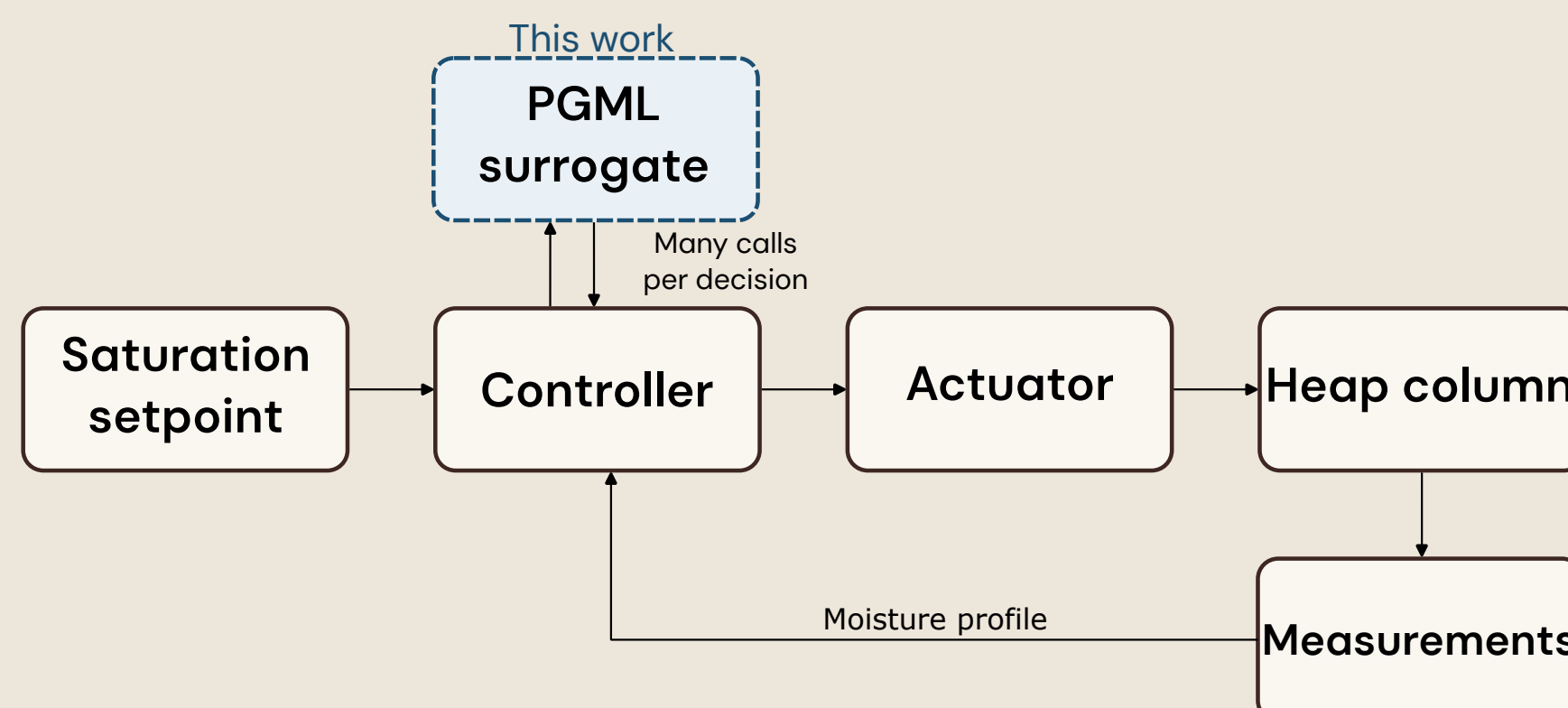


Figure 3: The NMPC control loop. Each decision fits the model then optimises irrigation; the predictor (this work, blue) replaces the fitting step that bottlenecks real-time use.

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References

- 1 IEA. Global Critical Minerals Outlook 2025. [iea.org/reports/global-critical-minerals-outlook-2025](https://www.iea.org/reports/global-critical-minerals-outlook-2025).
- 2 Marsden & Botz. Miner. Metall. Process. 2017. doi:10.19150/mmp.7505
- 3 Calvo et al. Resources 2016. doi:10.3390/resources5040036
- 4 McBride et al. Minerals 2018. doi:10.3390/min8010009

- 5 Liu & Hashemzadeh. Hydrometallurgy 2017. doi:10.1016/j.hydromet.2017.01.007
- 6 Olivares et al. Miner. Eng. 2025. doi:10.1016/j.mineng.2025.109346
- 7 Estay et al. Minerals 2023. doi:10.3390/min13060788
- 8 Karniadakis et al. Nat. Rev. Phys. 2021. doi:10.1038/s42254-021-00314-5